Here's How You Can Choose the Right Database For Your Business!

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Choosing the right database is becoming increasingly crucial for any business that involves large volumes of data. Of course, with a number of options, both in the proprietary and open source space, IT managers are spoilt for choice. One of the biggest challenges they face is choosing a database to suit their requirements at the most affordable price. Obviously, open source databases emerge as clear winners when it comes to weighing the burden on a company's budget.

Choosing a database is a long term decision and changing that decision later can be a difficult and expensive proposition. So, IT managers cannot afford to go wrong in the first instance. MySQL and PostgreSQL are the two popular alternatives in the world of open source databases, and are the most popular amongst admins preparing a new rollout.

According to Rohit Rai, sales director, India, South East Asia and Australia at EnterpriseDB, "The database market today is very mature and it is a mature technology too. We are talking about a 25 year-old database market. There are many databases that have come and gone, and now there are a few commercial choices that are left in the world. If you look at the database as a technology, apart from being pretty mature, it is standards compliant and well-documented. Open source databases also started out a long time back. If you look at the history of PostgreSQL, it's over 15 years old and has a healthy community around it."

Things to consider while selecting an open source database

With a number of database options available in the market, the job of IT managers actually becomes difficult while trying to choose one of them. So, how should admins go about it? The best point to start from is evaluating your requirements. A database needs to be selected based on the underlying data and the usage of that data. Some of the pointers are given below:

- What type of data is it? [Words (English/multilingual), numbers, books, images, movies, maps, etc]
- What is the volume of data that the database needs to contain? [Megabytes, terabytes or petabytes]
- How fast is the data generated? [Bytes/sec, MBps or GBps]
- How many people access it concurrently? [In the 10s, 100s, 1000s or millions]
- What is the read vs write volume? [Will only the same person who is writing read it, will one person write and millions read it, or will there be millions of writes but very few reads, i.e., log records?]
- Does the data need to be distributed? [Single computer or across countries]
- How much computation is involved in extracting information from the data? [Nothing - data is extracted as-is; some simple computation; complex joins of data or huge analytics computations]
- What expertise level do you expect your engineers to have? [Will they be able to manipulate data themselves or will you need expert database admins?]
- In what language does the data need to be accessed? [C, C++ to Ruby, Python or Golang]
- How valuable is the data? [Useless once the connection is lost, i.e., mobile phone locations or Web transactions; extremely important like bank accountsl; or critical,